



SniperSheet

AI-Powered Excel Formula Engine

Natural Language → Precise Excel Formulas in Seconds

35+
Formula Patterns

4
Smart Tabs

100%
Free AI

3
AI Models

Smart Hub

Commands

Cell Dimensions

Advanced Tools

100% Free

Bilingual



What is SniperSheet?

Complete product overview and key capabilities

SniperSheet is a professional Microsoft Excel task pane add-in that transforms natural language into precise, ready-to-use Excel formulas. Powered by Groq AI (Llama 3.3 70B — completely free), it supports both English and Arabic input, routes all AI calls through a US Replit server (no VPN needed anywhere), and falls back to a built-in local formula engine when offline.

Groq AI Engine
Llama 3.3 70B — free
No credit card required

Bilingual Support
English & Arabic input
Full RTL layout support

Proxy Architecture
US server relays AI calls
No VPN needed — worldwide

Local Fallback
35+ instant patterns
Zero latency — works offline

Office.js Integration
ReadWriteDocument access
Custom Excel Ribbon tab

Responsive Design
400px task pane optimized
Smooth scrolling UX

Describe your calculation in plain English or Arabic — SniperSheet generates the exact formula with explanation, confidence score, and formatting hints instantly.



Who is SniperSheet for?

Target users and primary use cases

Engineers
Complex engineering formulas in seconds

Financial Analysts
Data analysis and KPI tracking

Project Managers
Progress tracking and budgeting

Students
Statistical analysis and research

Smart Hub — AI Formula Engine

Tab 1 · Natural language → Excel formula via Groq AI

The core feature of SniperSheet. Type any description of your calculation in English or Arabic — such as "calculate overtime for hours over 40 at 1.5x rate" — then click "Smart Analysis". The AI engine processes your request via Groq AI and returns the exact formula with a detailed explanation, confidence score (0–100%), and optional formatting hints.

Full Feature List

Everything Smart Hub offers

- AI formula generation via Groq (Llama 3.3 70B)
- Confidence scoring: 0–100% per formula
- Formula type auto-detection (IF, VLOOKUP, SUM...)
- Word Radar: detects typos & ambiguous terms
- Style hints: cell colors, bold, italic formatting
- Full history log with timestamps
- Example prompts library in English & Arabic
- Keyboard shortcut for instant analysis
- Local formula engine — works fully offline
- 3-model AI cascade with auto-retry on failure
- Status indicators: success, warning, error
- Detailed step-by-step formula explanation
- One-click formula copy to clipboard
- One-click apply formula directly to Excel cell
- Supports deeply nested complex formulas
- Smart error handling with fix suggestions

How Smart Hub Works

Step-by-step from description to formula

1

Type

Describe your calculation in plain English or Arabic in the input field

2

Analyze

Click "Smart Analysis" or use the keyboard shortcut to trigger processing

3

Process

Request is securely routed through Replit server to Groq AI for analysis

4

Result

Formula appears with confidence score, explanation, and formatting hints

Commands — Formula Aliases

Tab 2 · Arabic & English names for instant calculations

Type the function name in English or Arabic with a cell range to get an instant result. For example, type "SUM B1:B10" or the Arabic equivalent "جمع" with a range. The Commands tab includes a searchable reference table of all available function names with Arabic aliases and usage examples.

Command Name	Description	Formula Generated
SUM / جمع	Sum of a cell range	=SUM(B1:B10)
MULTIPLY / ضرب	Product of values	=PRODUCT(A1:A5)
AVERAGE / متوسط	Arithmetic mean	=AVERAGE(D1:D20)
MAX / أكبر	Maximum value	=MAX(C1:C50)
MIN / أصغر	Minimum value	=MIN(C1:C50)
COUNT / عدد	Count numeric cells	=COUNT(E1:E100)
BONUS / مكافأة	Adds 15% bonus automatically	=A1*1.15
TAX / ضريبة	Calculates 15% tax	=A1*0.15
PERCENTAGE / نسبة	Percentage between two values	=(A1/B1)*100
IF / إذا	Simple conditional logic	=IF(A1>1000, "High", "Low")

Tip: You can combine commands — for example type "BONUS SUM B1:B10" to sum a range and then apply the 15% bonus automatically. Use the in-app searchable reference table to discover all available commands.



Cell Dimensions

Tab 3 · Calculate optimal width & height for any cell content

Calculates the optimal width and height for an Excel cell based on its content, accounting for Arabic character width, RTL text direction, font size, bold and italic styling, and padding margins. Supports both single-cell and batch processing modes.

- Arabic text character width calculation
 - RTL text direction awareness
 - Font size & bold/italic adjustment
 - Single cell & batch processing modes
 - Pixel and Excel unit output
 - One-click copy results to clipboard
- Mixed Arabic + English text support
 - Switch between measurement units
 - Preview dimensions before applying
 - Apply dimensions directly to Excel
 - Save favorite settings
 - Intuitive easy-to-use interface



Advanced Tools

Tab 4 · Three professional productivity utilities

Empty Field Radar

Scan CSV or tab-separated data for missing values. Get a color-coded visual grid plus a precise list of every empty cell coordinate.

- Paste CSV or tab-separated data
- Color grid: filled=green, empty=red
- Exact empty cell coordinate list
- Auto header row detection
- Filled vs empty summary count

Smart Print-Fit

Calculate optimal column widths and font sizes to fit your spreadsheet onto A4 or A3 paper — eliminating all trial-and-error print formatting.

- A4 & A3 paper size support
- Portrait & landscape orientation
- Excel column width units output
- Recommended font size output
- Print-ready result in 3 clicks

Professional Report

Transform raw pasted data into a styled HTML table with alternating row colors and bold column headers, then send directly to the browser print dialog.

- Styled HTML from raw data
- Alternating row color styling
- Bold column headers
- Auto header row detection
- One-click print dialog trigger

AI Proxy Architecture

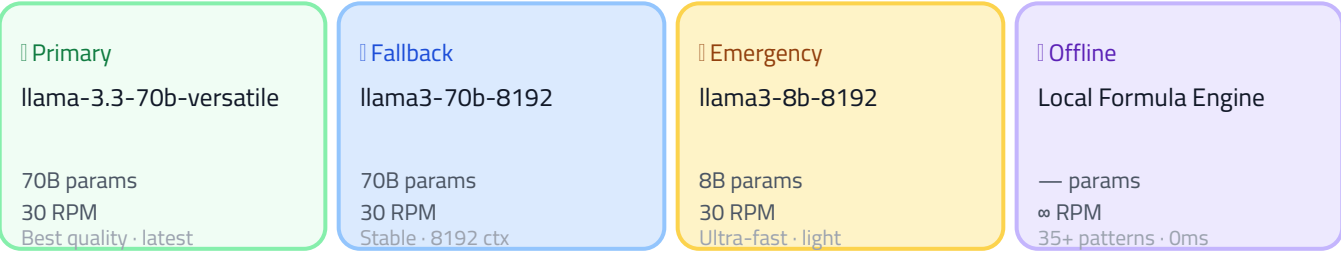
How SniperSheet connects to AI without exposing API keys



Security: GROQ_API_KEY is stored as a server-side secret on Replit. Users everywhere — including Iraq — connect only to Replit's HTTPS endpoint. They never interact with Groq directly. Zero VPN required.

AI Model Cascade

3 free Groq models + offline local fallback — auto-retry



Full Technical Specifications

Complete add-in configuration and stack details

Office.js Manifest v1.1 Add-in Type Task Pane Pane Width 400px Permission Level ReadWriteDocument Default Locale ar-SA (Arabic) Ribbon Tab Custom SniperSheet	Frontend Stack React 18 + Vite + TypeScript Backend Stack Express.js + Node.js 24 AI Library Groq SDK (Official) Package Manager pnpm Monorepo Workspace Hosting Replit Autoscale Deploy Styling Tailwind CSS + shadcn/ui
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Real Formula Examples

Generated by Smart Hub AI from natural language input

Natural Language Input	Category	Formula Generated
Calculate overtime for hours over 40 at 1.5x rate	Conditional	=IF(A1>40, (A1-40)*1.5, 0)
Grade student: A+ above 90, B above 75, else C	Conditional	=IFS(B1>=90, "A+", B1>=75, "B", "C")
Find value in column A, return matching from column B	Lookup	=XLOOKUP(D1, A:A, B:B, "Not found")
Sum all sales amounts greater than 1000	Statistical	=SUMIF(C:C, ">1000", D:D)
Monthly payment for 30-year loan at 5% annual	Financial	=PMT(0.05/12, 360, A1)
Calculate person age in years from birth date in A1	Date	=DATEDIF(A1, TODAY(), "Y")
Rank employee by highest sales figure descending	Statistical	=RANK(A1, A:A, 0)
Add 15% bonus if sales exceed 10,000	Conditional	=IF(A1>10000, A1*1.15, A1)
Count students who scored 60 or higher	Statistical	=COUNTIF(B:B, ">=60")

Confidence: Green > 85% = high accuracy · Orange 50–85% = acceptable, review prompt · Red < 50% = revise with more specific details. Clearer prompts consistently yield higher confidence scores.

Example Prompts That Work Well

Best practices for writing effective formula requests

Specific & clear

Calculate overtime pay: hours worked minus 40, multiplied by 1.5 times hourly rate in column B

Name your columns

Look up the employee ID in column A and return their department name from column D

Include conditions

If sales in column C exceed 10,000, apply 15% bonus, otherwise return the original value

Too vague

Calculate something with numbers — rephrase with specific context and column references

Installation Guide

4 simple steps to load SniperSheet in Microsoft Excel

1

Download manifest.xml

Navigate to the manifest URL provided by your administrator. The manifest.xml file downloads automatically to your device.

2

Open Excel Add-ins

In Microsoft Excel: click Insert → Add-ins → My Add-ins → Manage My Add-ins (bottom of the dropdown).

3

Upload manifest.xml

Click "Upload My Add-in", select the downloaded manifest.xml file. The SniperSheet ribbon tab appears automatically.

4

Open Smart Hub

Click the SniperSheet tab in the Excel Ribbon → click "Open Sniper Hub". The task pane opens on the right side!

Technology Stack

Complete stack used to build SniperSheet

■

Frontend

React 18 + Vite + TypeScript

■

Backend

Express.js + Node.js 24

■

AI SDK

Groq SDK — Llama 3.3 70B

■

Styling

Tailwind CSS + shadcn/ui

■

Package Mgr

pnpm Monorepo Workspace

■

Office.js

Manifest v1.1 Task Pane

■

Permission

ReadWriteDocument

■

Hosting

Replit Autoscale Deployment

System Requirements

- Microsoft Excel 2016 or later (Windows or Mac)
- Internet connection for AI features (offline mode available)
- Modern browser (Chrome / Edge / Firefox) — add-in runs in browser context

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SniperSheet v1.0 · Official English Guide · Built on Replit · Powered by Groq AI (Free Tier)



Developer Profile
Creator of SniperSheet — Mustafa Alsahlany

Mustafa Alsahlany

Developer & Designer — SniperSheet Excel Add-in
Built with React, Express.js, TypeScript, and Groq AI on Replit
Iraq — 2026

- Replit — Active
- Groq AI — Free
- v1.0.0
- April 2026



Frequently Asked Questions
Common questions about SniperSheet

Is SniperSheet truly 100% free?

Yes. SniperSheet uses Groq's free tier (Llama 3.3 70B, 30 requests/min). The API key is stored on the developer's server. You never need to create a Groq account or pay anything.

What happens if the AI is unavailable?

SniperSheet has a 3-model cascade: if the primary model fails, it tries two fallbacks automatically. If all cloud models fail, the local formula engine activates with 35+ instant patterns.

What Excel versions are supported?

Excel 2016 and later (Windows and Mac). The add-in uses Office.js Manifest v1.1 with ReadWriteDocument permission level.

Do users in Iraq need a VPN?

No. All AI requests route through a US-based Replit server. Users connect to Replit's HTTPS endpoint, not directly to Groq. No VPN, proxy, or workaround is needed.

Does it work without internet?

Yes — the local formula engine handles SUM, IF, VLOOKUP, and 35+ other patterns offline with zero latency. AI features require an internet connection.

Can I use Arabic to generate formulas?

Absolutely. Arabic input is fully supported. Type your description in Arabic and SniperSheet generates the exact Excel formula with an Arabic explanation.